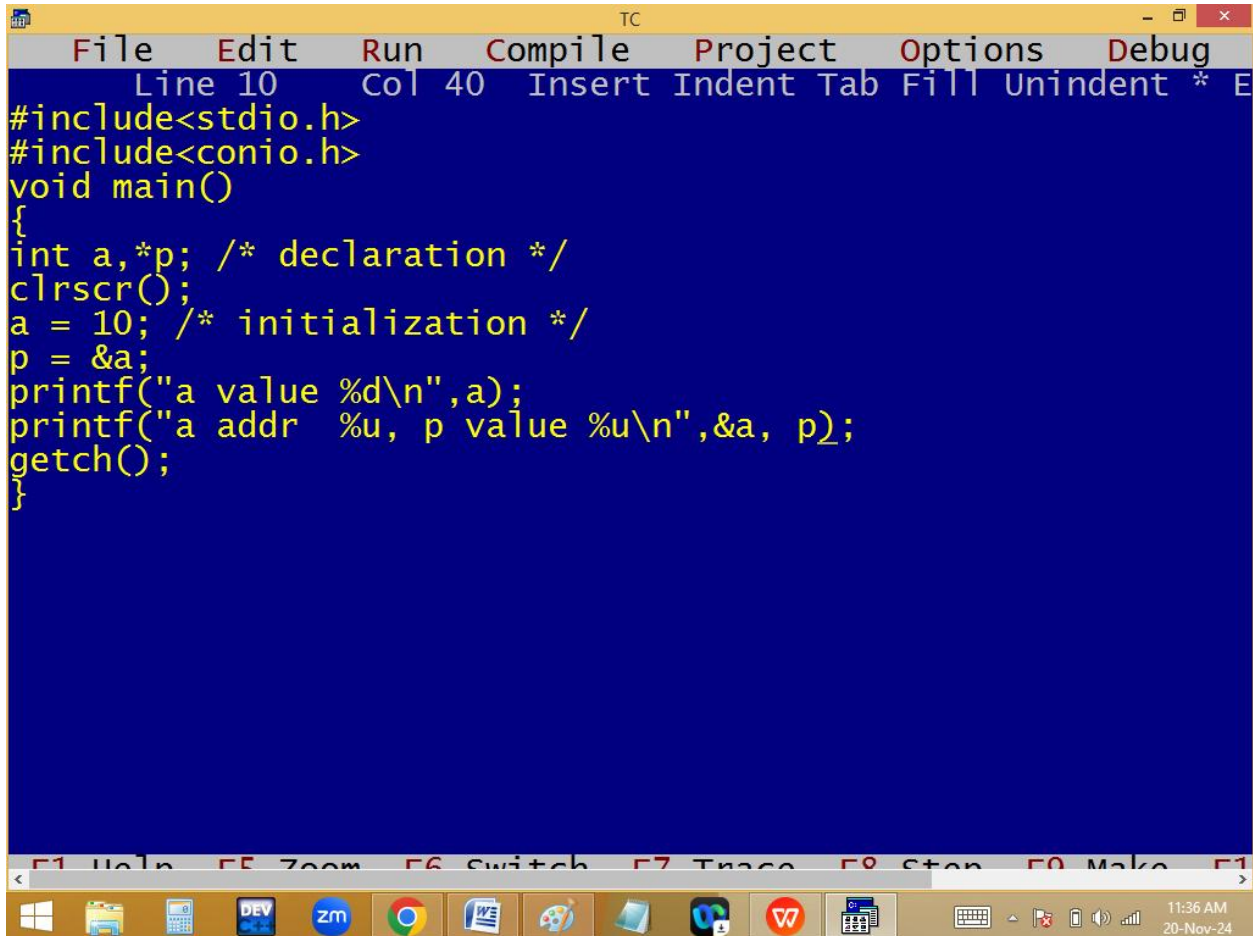
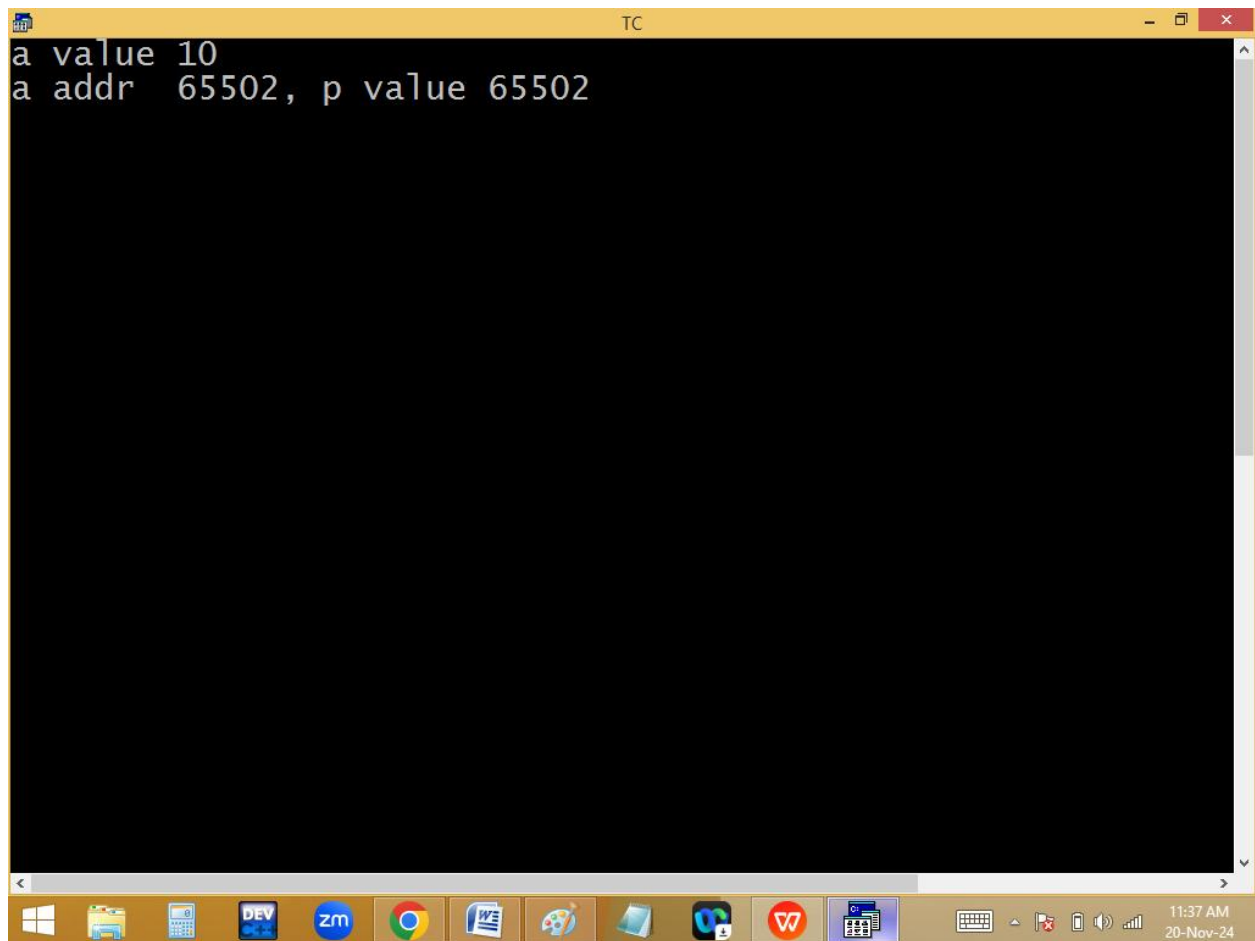


Address operators:

1. & - address of the variable
2. * - address of another variable [pointer]



```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 40 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a,*p; /* declaration */
clrscr();
a = 10; /* initialization */
p = &a;
printf("a value %d\n",a);
printf("a addr %u, p value %u\n",&a, p);
getch();
}
```



The image shows a screenshot of a Turbo C++ (TC) window. The window has a yellow title bar with the text "TC" and standard window controls. The main area is black with white text. The text displayed is:

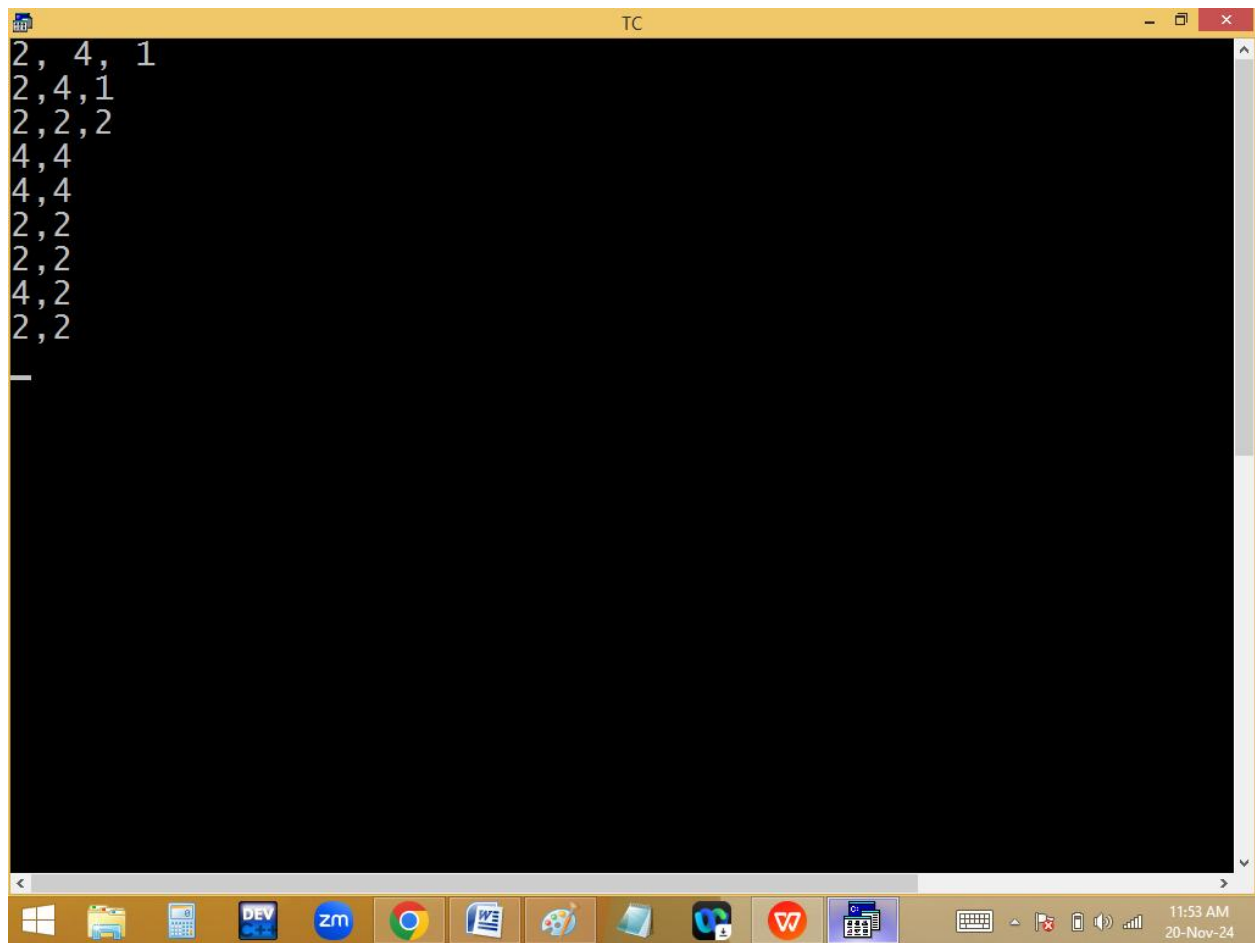
```
a value 10  
a addr 65502, p value 65502
```

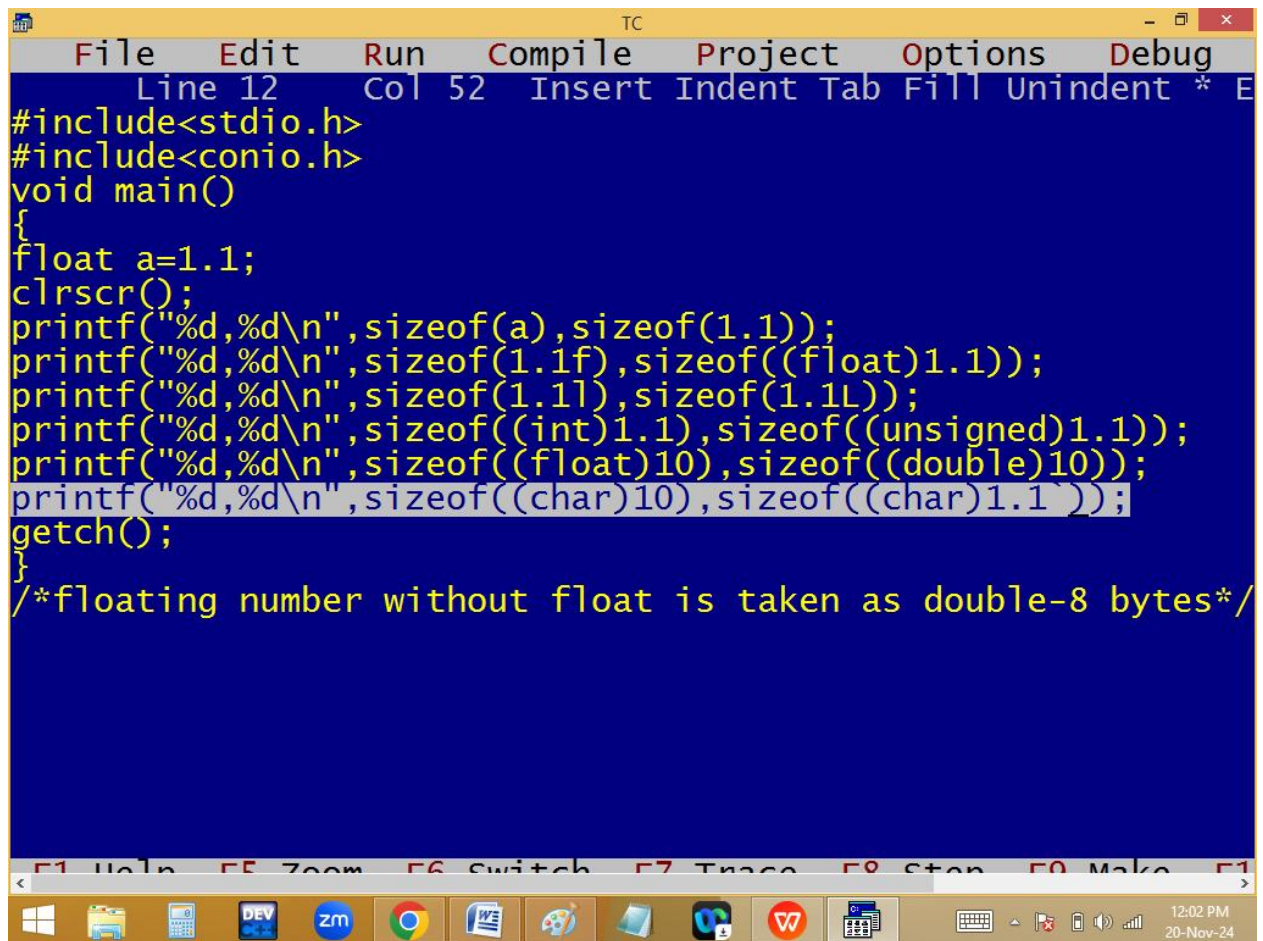
The window is running on a Windows operating system, as evidenced by the taskbar at the bottom. The taskbar includes icons for various applications like File Explorer, Calculator, DEV, zm, Chrome, Word, and others. The system clock in the bottom right corner shows "11:37 AM" and "20-Nov-24".

Sizeof() operator:

It return the no of bytes taken by a variable / data type / value.

```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 51 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
float b;
char c;
clrscr();
printf("%d, %d, %d\n", sizeof(a), sizeof(b), sizeof(c));
printf("%d,%d,%d\n", sizeof(int), sizeof(float), sizeof(char));
printf("%d,%d,%d\n", sizeof(-5), sizeof(5), sizeof(32767));
printf("%d,%d\n", sizeof(32768), sizeof(50000));
printf("%d,%d\n", sizeof(132768), sizeof(500000));
printf("%d,%d\n", sizeof(32768u), sizeof(50000U)); /*suffix*/
printf("%d,%d\n", sizeof((unsigned)32768), sizeof((int)50000));
printf("%d,%d\n", sizeof(100000u), sizeof((unsigned)100000));
printf("%d,%d\n", sizeof(50000u), sizeof((unsigned)50000));
/*prefix*/
getch();
}
/*After 32767 every int is long int - 4 bytes*/
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

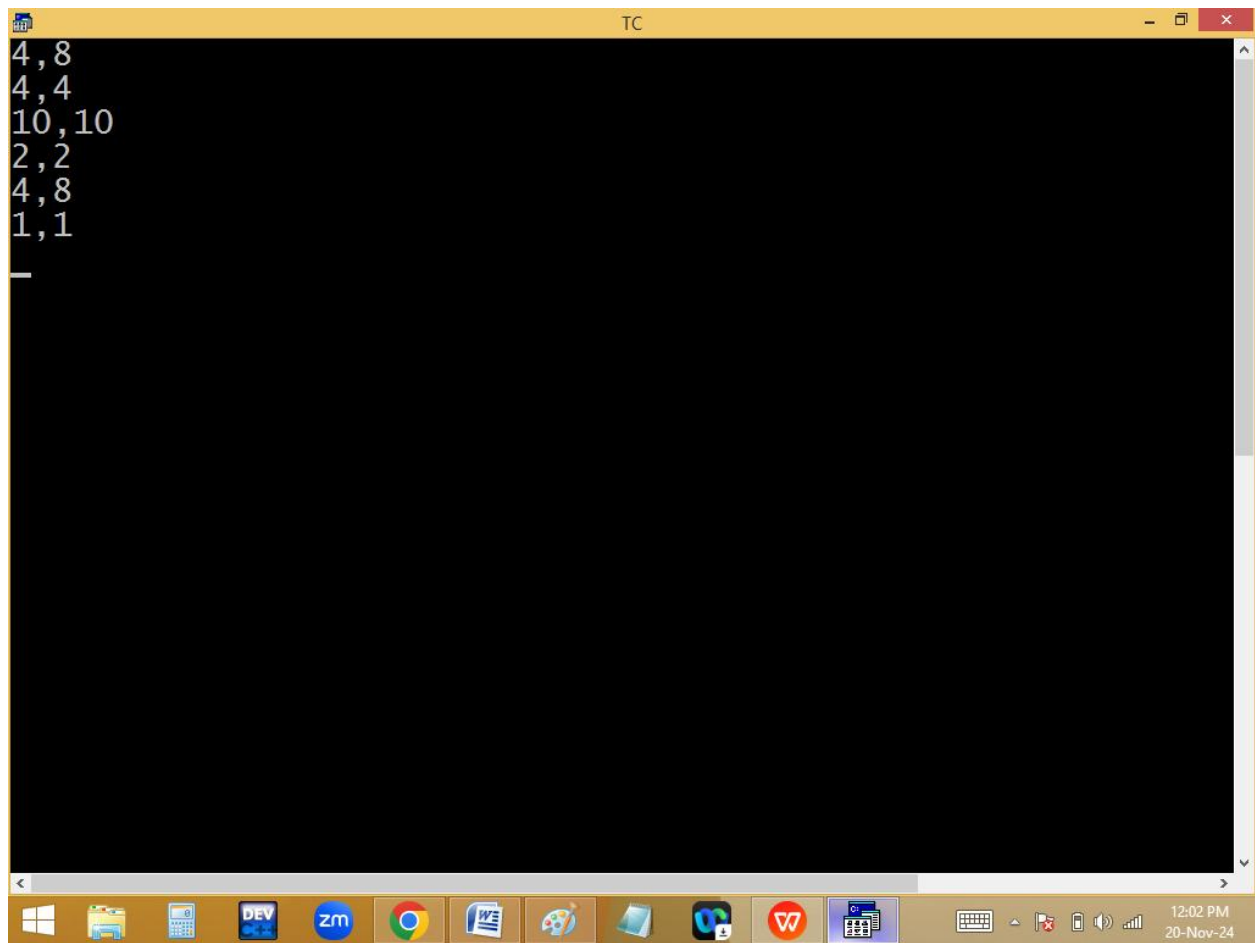




```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 52 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
float a=1.1;
clrscr();
printf("%d,%d\n",sizeof(a),sizeof(1.1));
printf("%d,%d\n",sizeof(1.1f),sizeof((float)1.1));
printf("%d,%d\n",sizeof(1.1l),sizeof(1.1L));
printf("%d,%d\n",sizeof((int)1.1),sizeof((unsigned)1.1));
printf("%d,%d\n",sizeof((float)10),sizeof((double)10));
printf("%d,%d\n",sizeof((char)10),sizeof((char)1.1));
getch();
}
/*floating number without float is taken as double-8 bytes*/
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:02 PM 20-Nov-24



```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
char s1[10]="abc", s2[]="abc";
clrscr();
printf("%d,%d,%d\n",sizeof(s1),sizeof(s2),sizeof("abc"));
printf("%d,%d\n",printf("kishore"),sizeof("kishore"));
printf("%d\n",sizeof("kishore naidu\n"));
printf("kishore addr is %u\n","kishore");
printf("%d, %d\n",sizeof("kishore")+1, sizeof("kishore"+1));
printf("%d, %d\n",sizeof("1.234"), sizeof(sizeof("abc")));
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:18 PM
20-Nov-24

```
TC
10,4,4
kishore7,8
15
kishore addr is 467
9, 2
6, 2
_
```

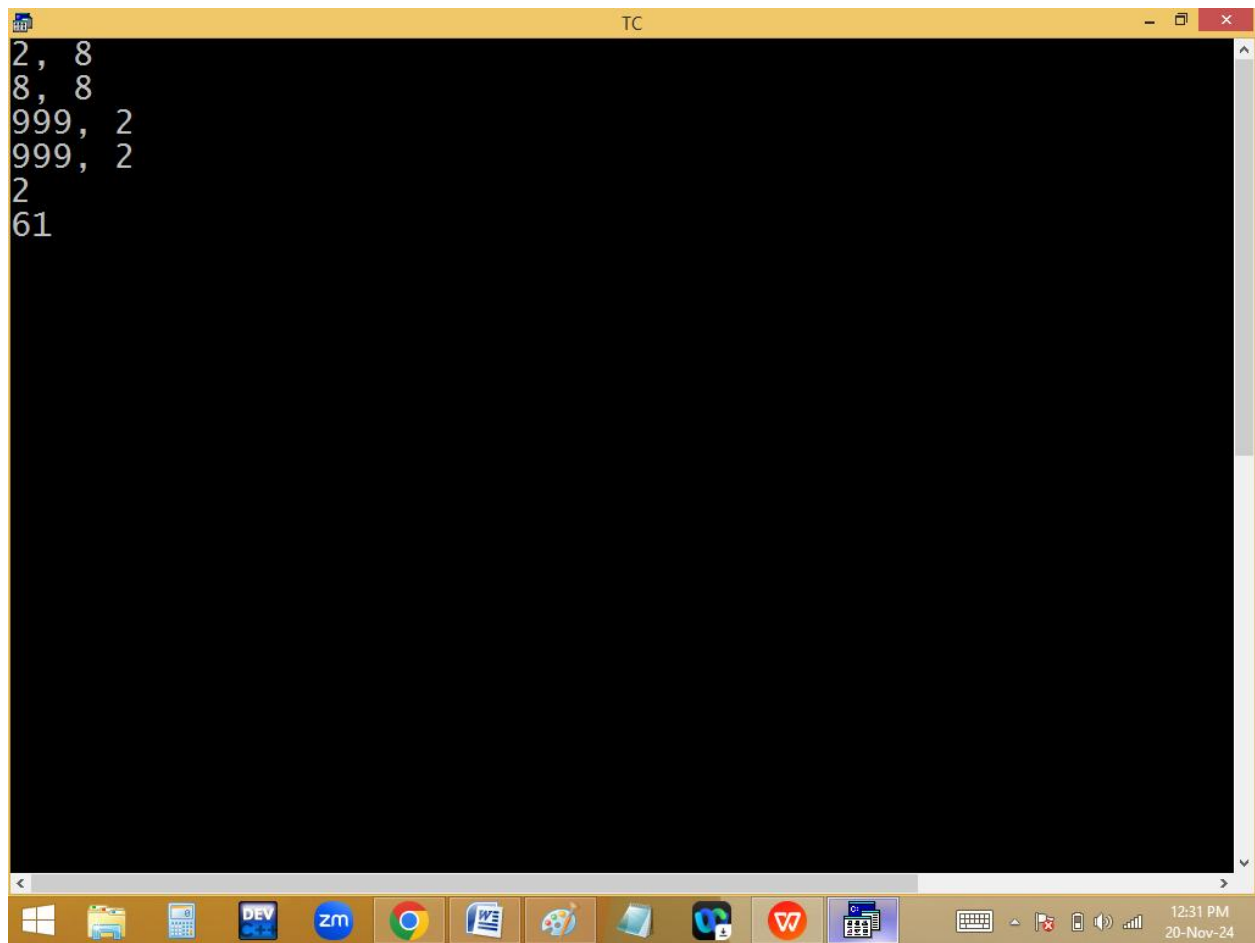
sizeof(sizeof("abc"))

→ (4)

2

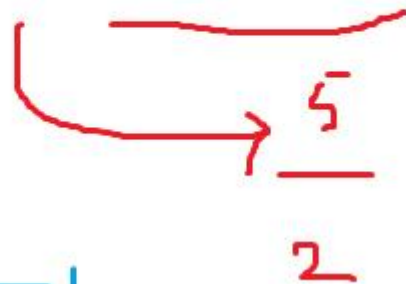
```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 16 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=999;
clrscr();
printf("%d, %d\n",sizeof(1.2, 34), sizeof(34, 1.2));
printf("%d, %d\n",sizeof(1.2+34), sizeof(34-1.2));
printf("%d, %d\n",a, sizeof(++a));
printf("%d, %d\n",a, sizeof(a=777));
printf("%d\n",sizeof(printf("Hello")));
printf("%d\n",printf("%d",sizeof("Hello")));
getch();
}
/* Note: In sizeof expressions not considered */

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

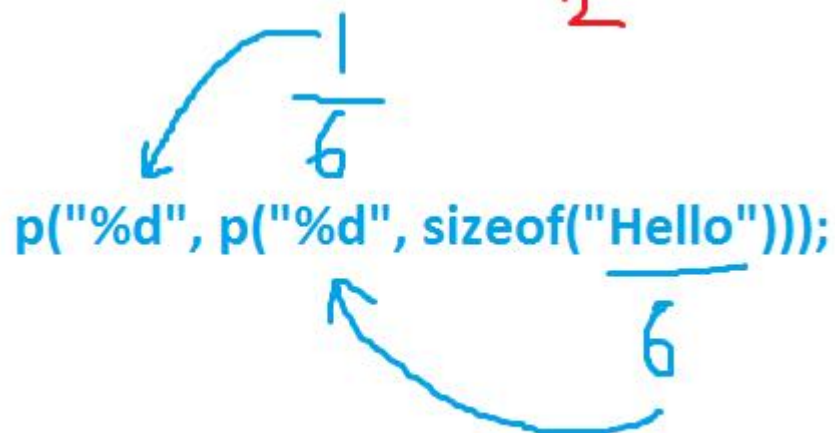


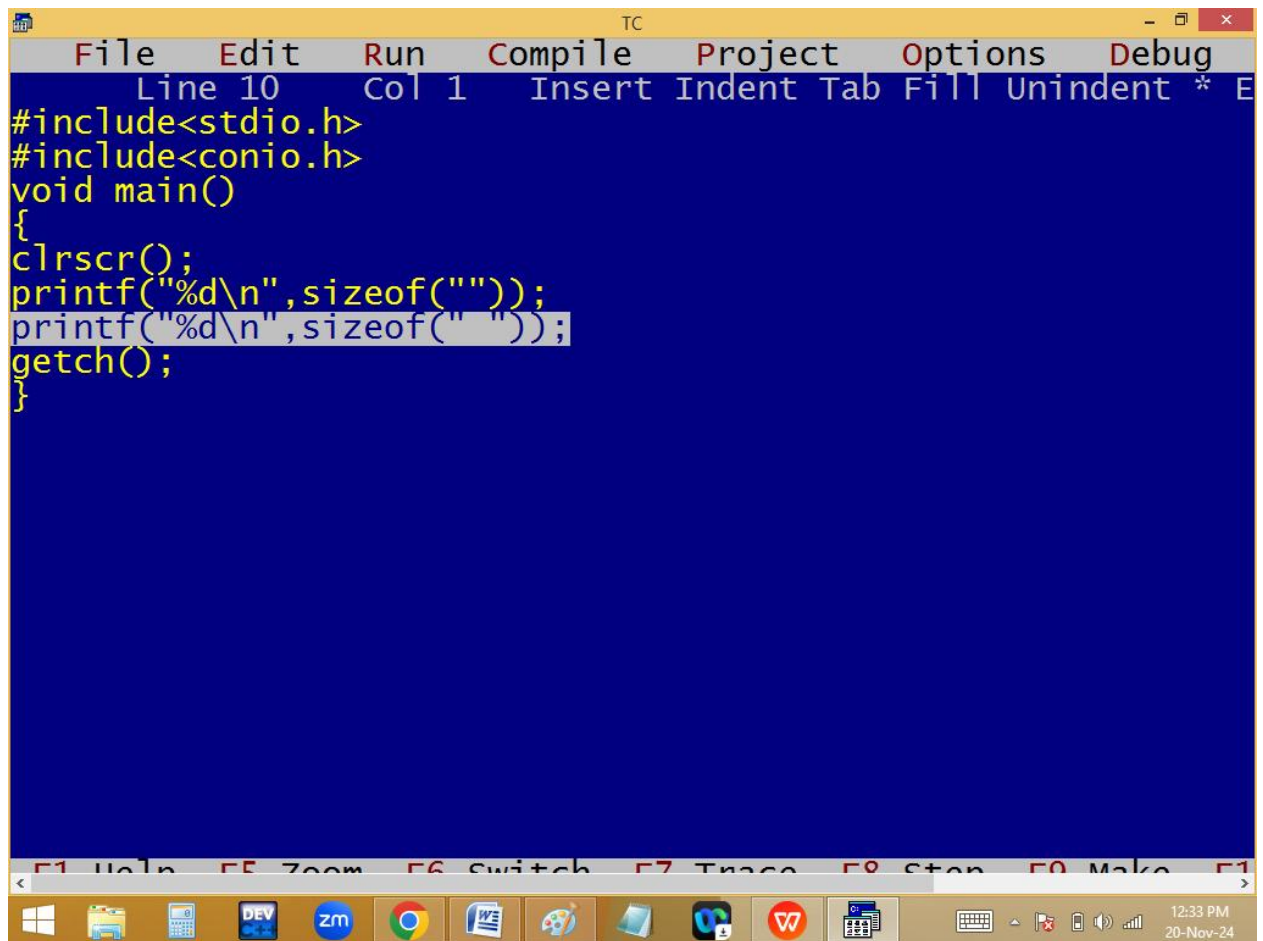
`p("%d", sizeof(printf("Hello")));`

5
2



6
p("%d", p("%d", sizeof("Hello")));

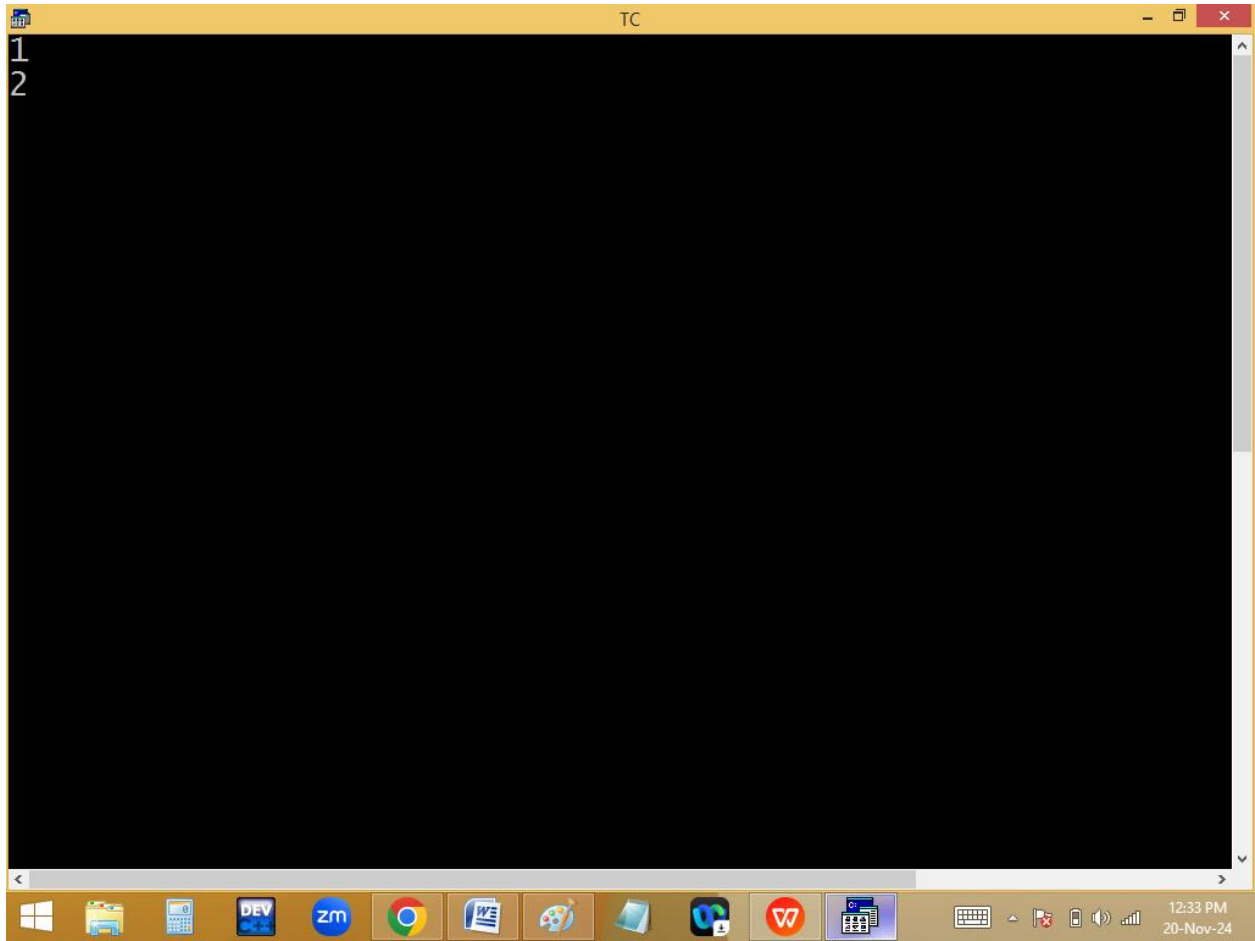


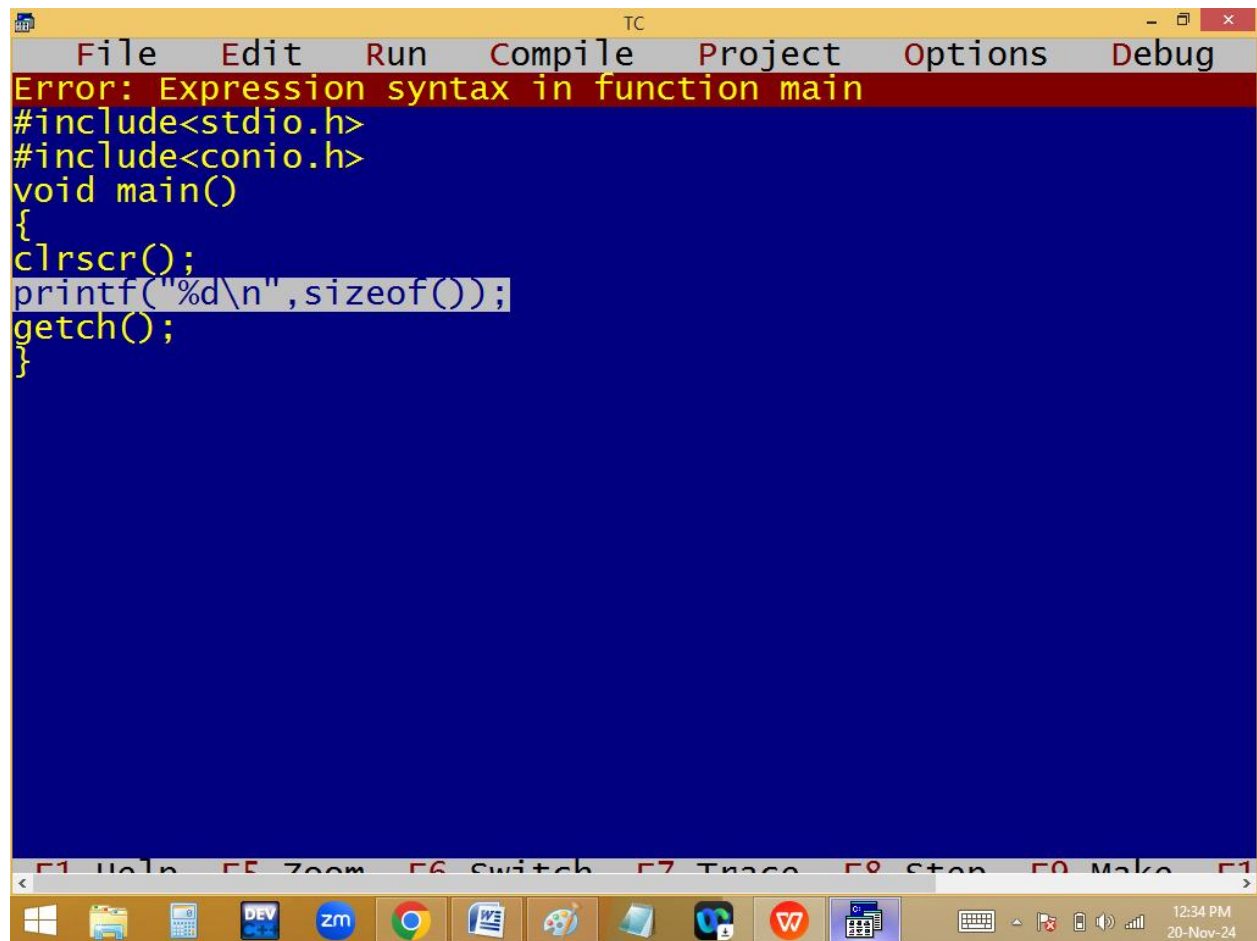


```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof(""));
printf("%d\n",sizeof(" "));
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:33 PM
20-Nov-24

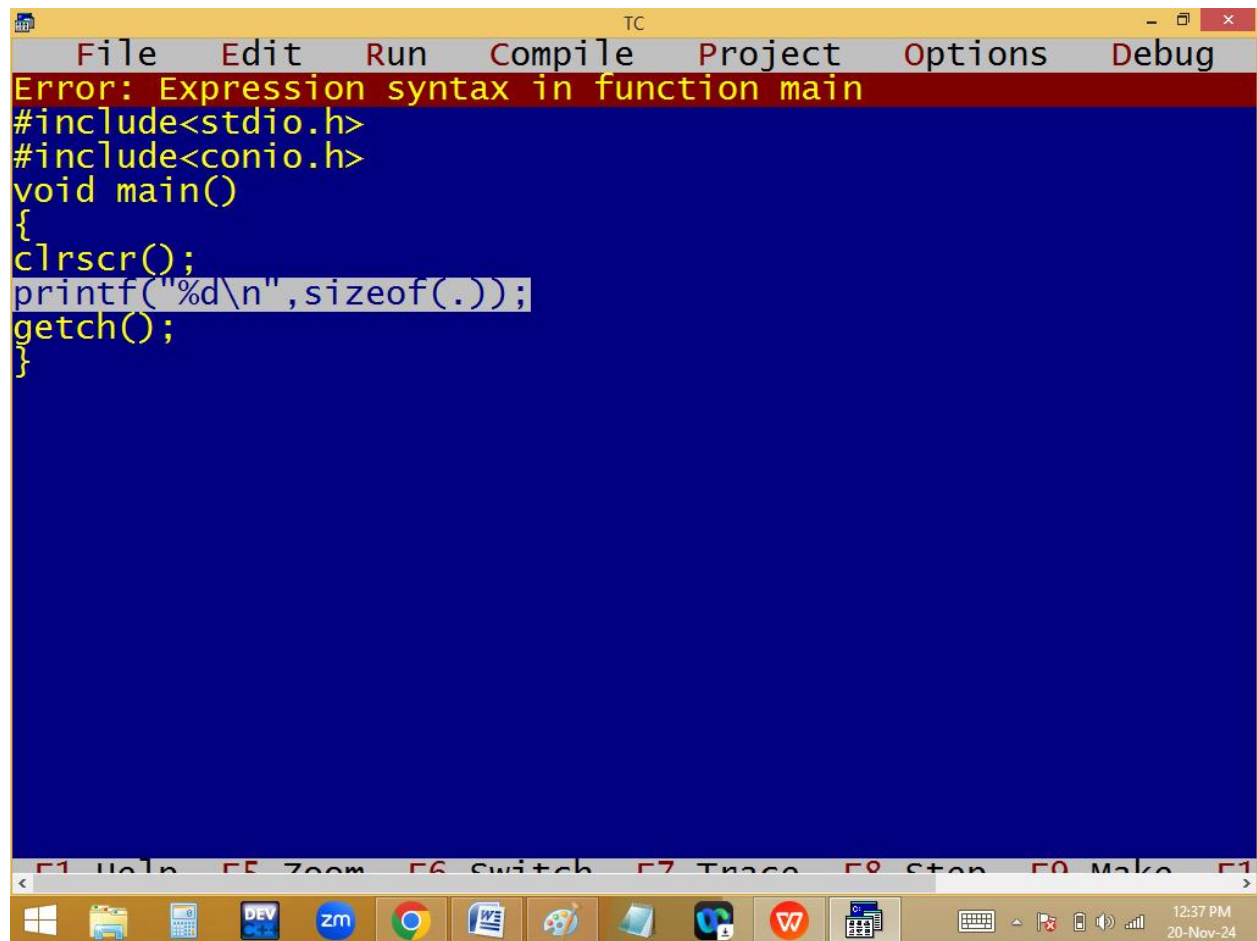




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top reads "Error: Expression syntax in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof());
getch();
}
```

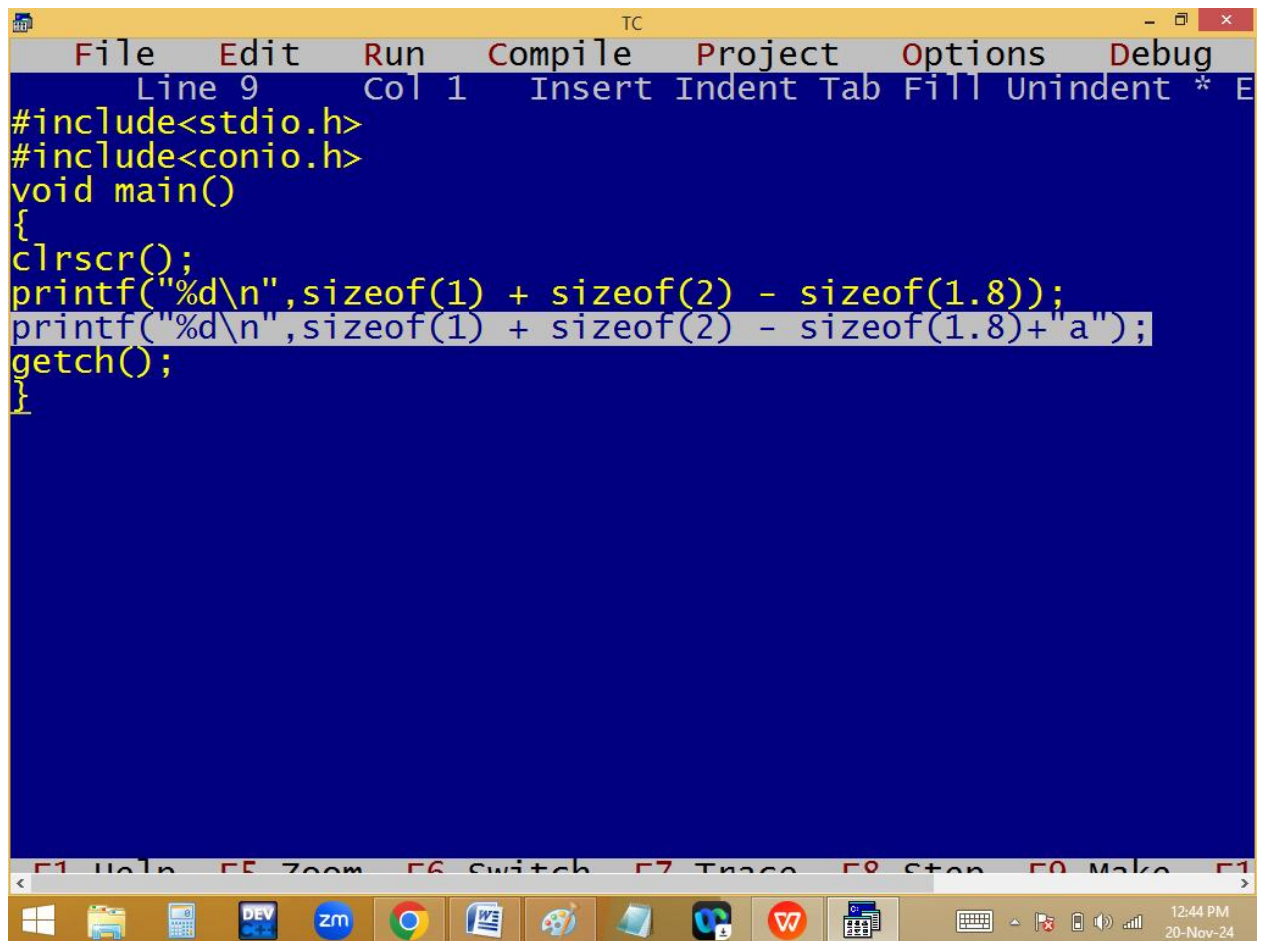
The line `printf("%d\n",sizeof());` is highlighted with a light blue selection. Below the code editor, a status bar shows function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar at the bottom includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, a folder, VS Code, and Word. System tray icons on the right show keyboard, volume, and network status. The system clock displays "12:34 PM" and "20-Nov-24".



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top reads "Error: Expression syntax in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof(.));
getch();
}
```

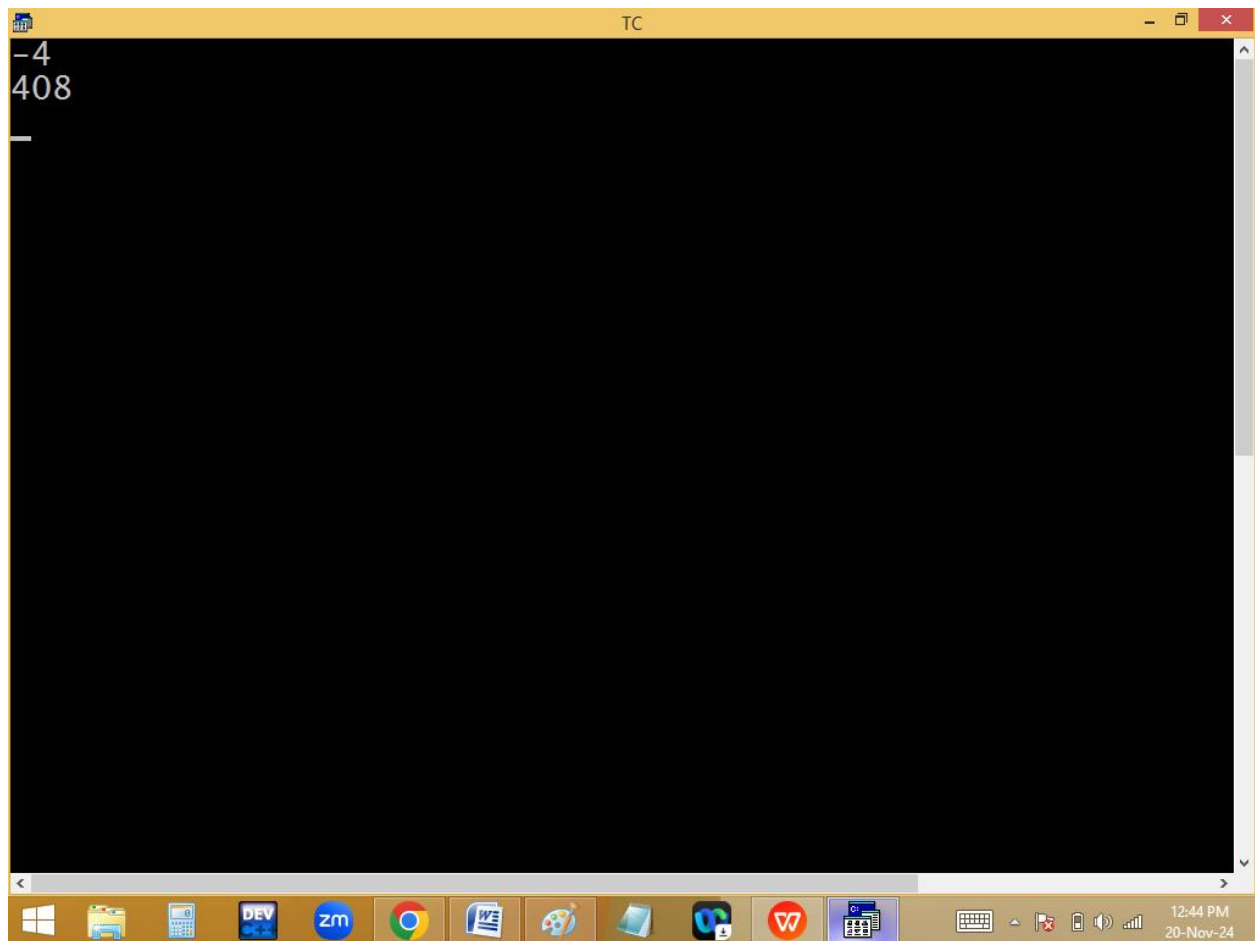
The line `printf("%d\n",sizeof(.));` is highlighted in light blue. Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, VS Code, and a red 'W' icon. The system clock in the bottom right corner shows "12:37 PM" and "20-Nov-24".



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 9 Col 1" and provides keyboard shortcuts for "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof(1) + sizeof(2) - sizeof(1.8));
printf("%d\n",sizeof(1) + sizeof(2) - sizeof(1.8)+"a");
getch();
}
```

The second `printf` statement is highlighted in grey. It contains a format specifier `%d` followed by a string literal `"a"`, which is a mismatch. The Windows taskbar at the bottom shows various application icons, including Windows Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, and others. The system clock in the bottom right corner displays "12:44 PM" and "20-Nov-24".



Here 408 is “a” addr – 4